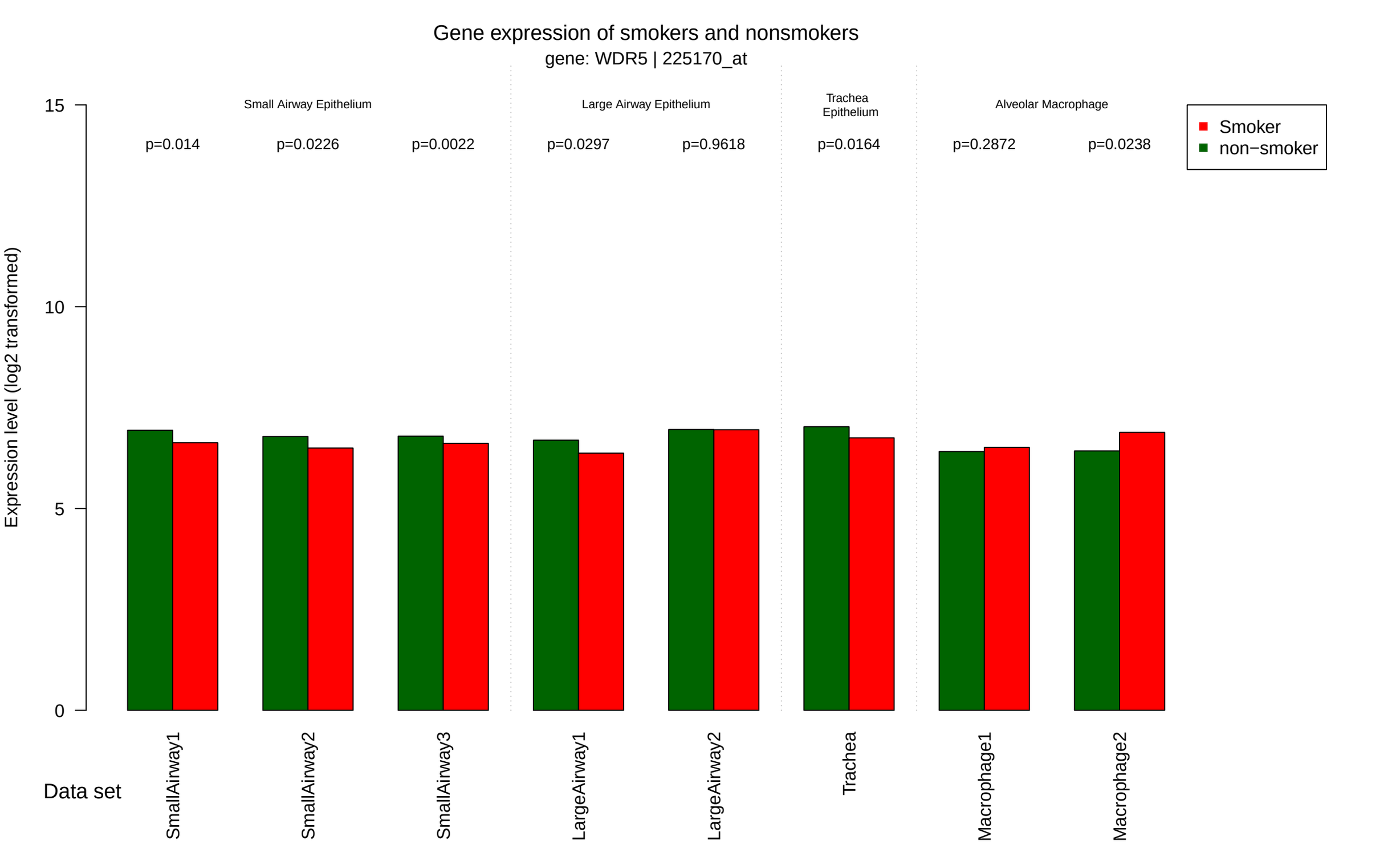
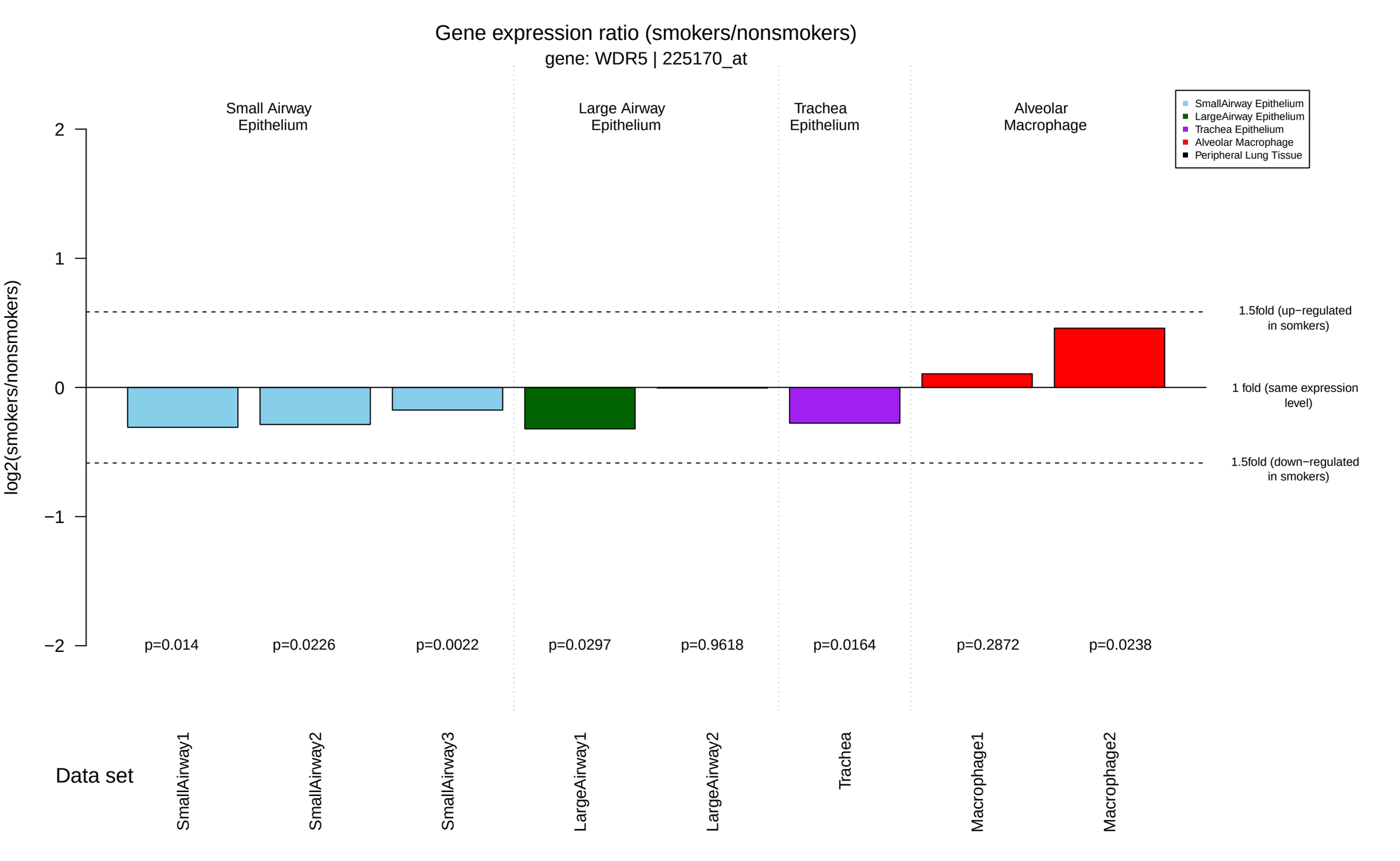
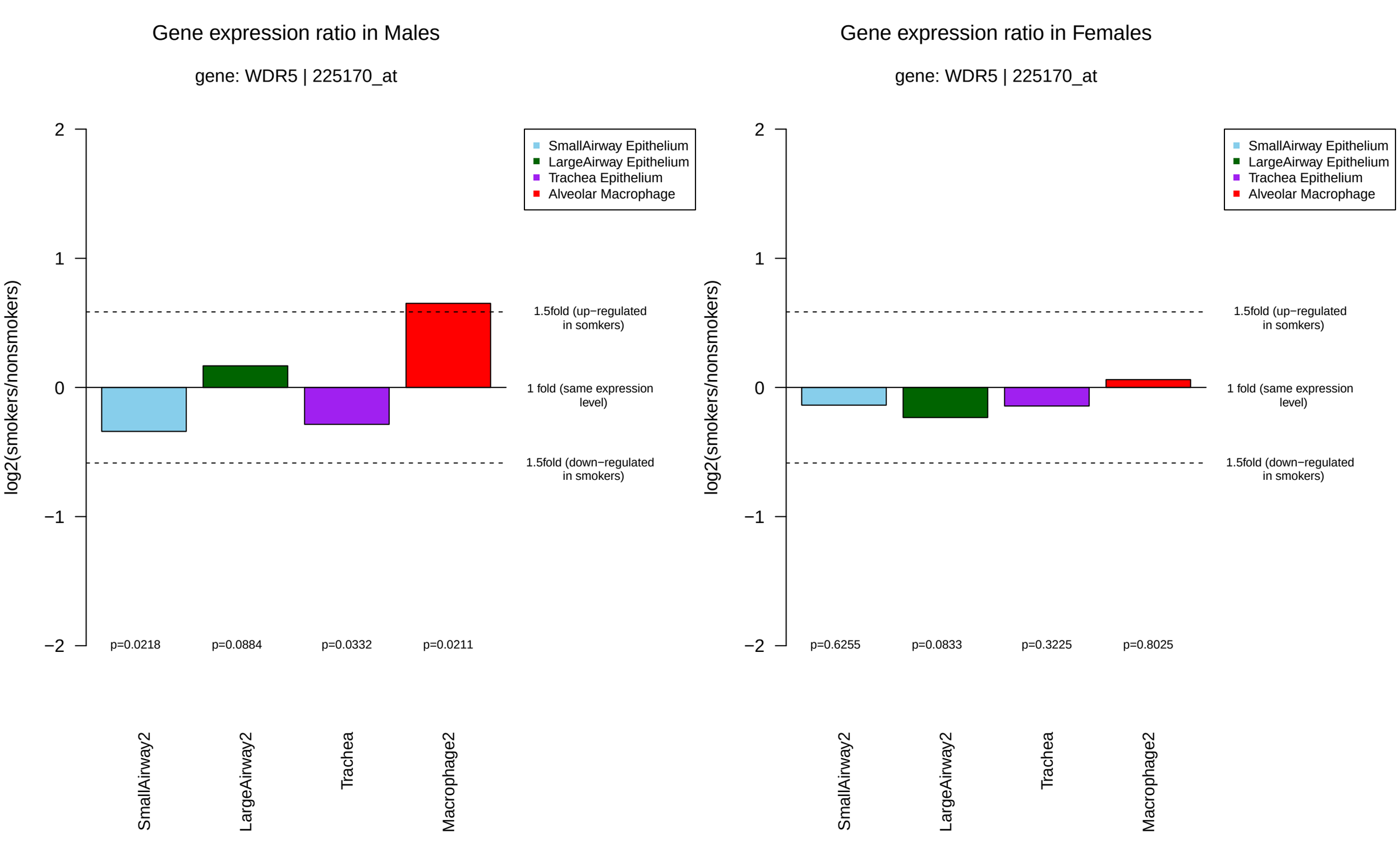
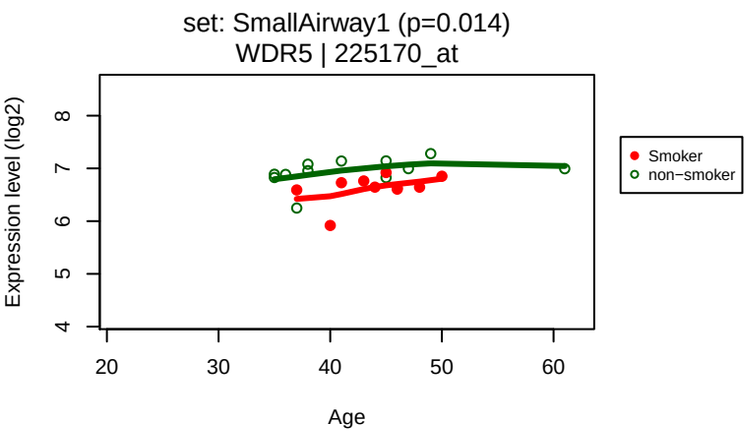
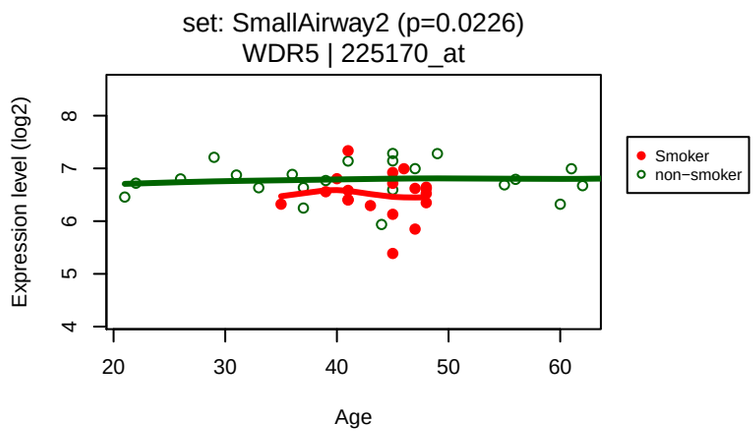
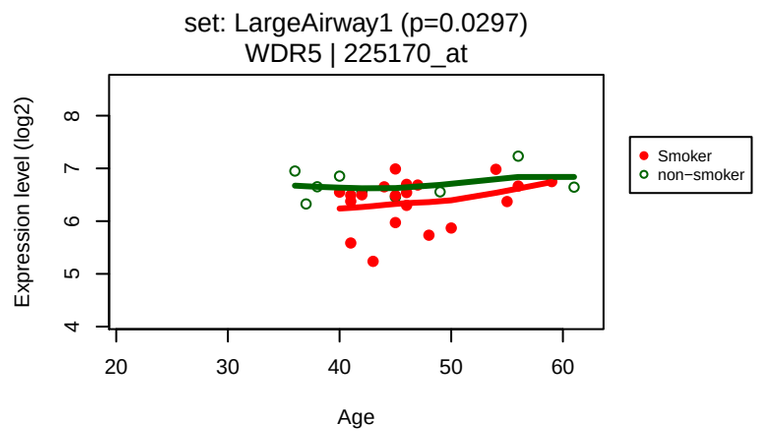

Gene expression of smokers and nonsmokers
gene: WDR5 | 225170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.014
p=0.0226
p=0.0022
p=0.0297
p=0.9618
p=0.0164
p=0.2872
p=0.0238
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

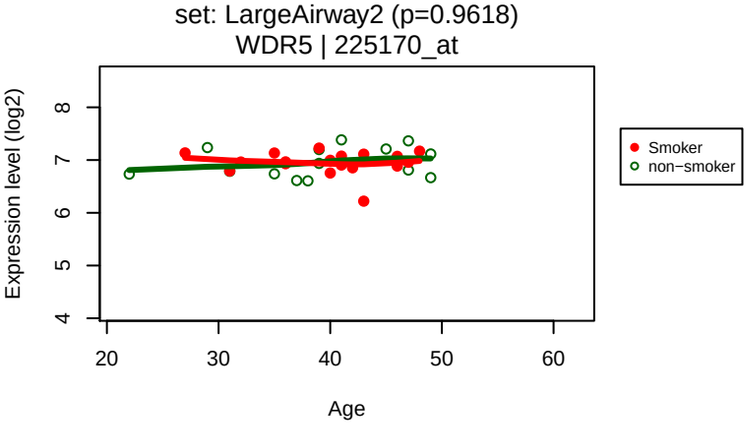

Gene expression ratio (smokers/nonsmokers)
gene: WDR5 | 225170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.014
p=0.0226
p=0.0022
p=0.0297
p=0.9618
p=0.0164
p=0.2872
p=0.0238
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

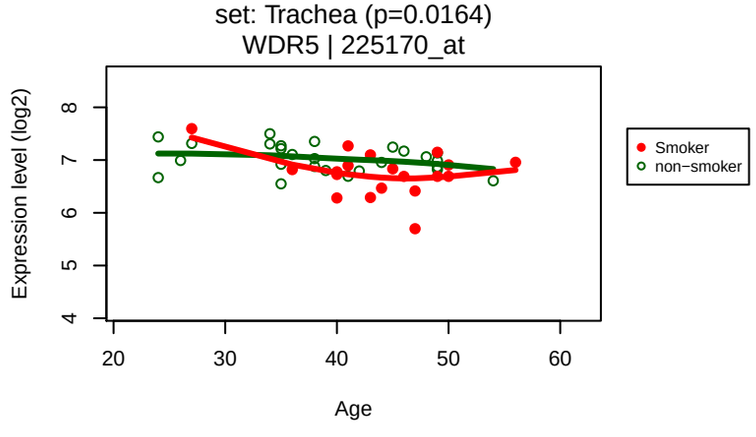

Gene expression ratio in Males
gene: WDR5 | 225170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0218
p=0.0884
p=0.0332
p=0.0211
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WDR5 | 225170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6255
p=0.0833
p=0.3225
p=0.8025
SmallAirway2
LargeAirway2
Trachea
Macrophage2

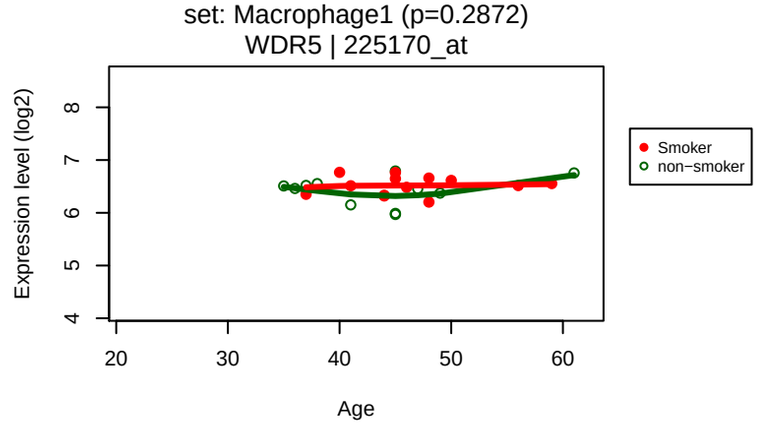

set: SmallAirway1 (p=0.014)
WDR5 | 225170_at
Expression level (log2)
Age
Smoker
non−smoker

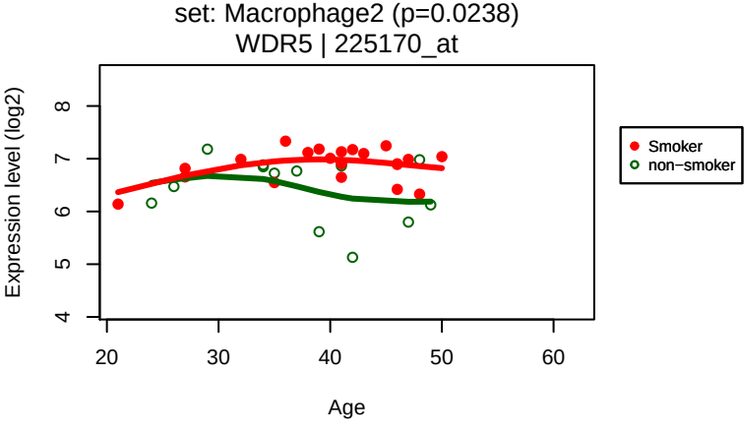

set: SmallAirway2 (p=0.0226)
WDR5 | 225170_at
Expression level (log2)
Age
Smoker
non−smoker

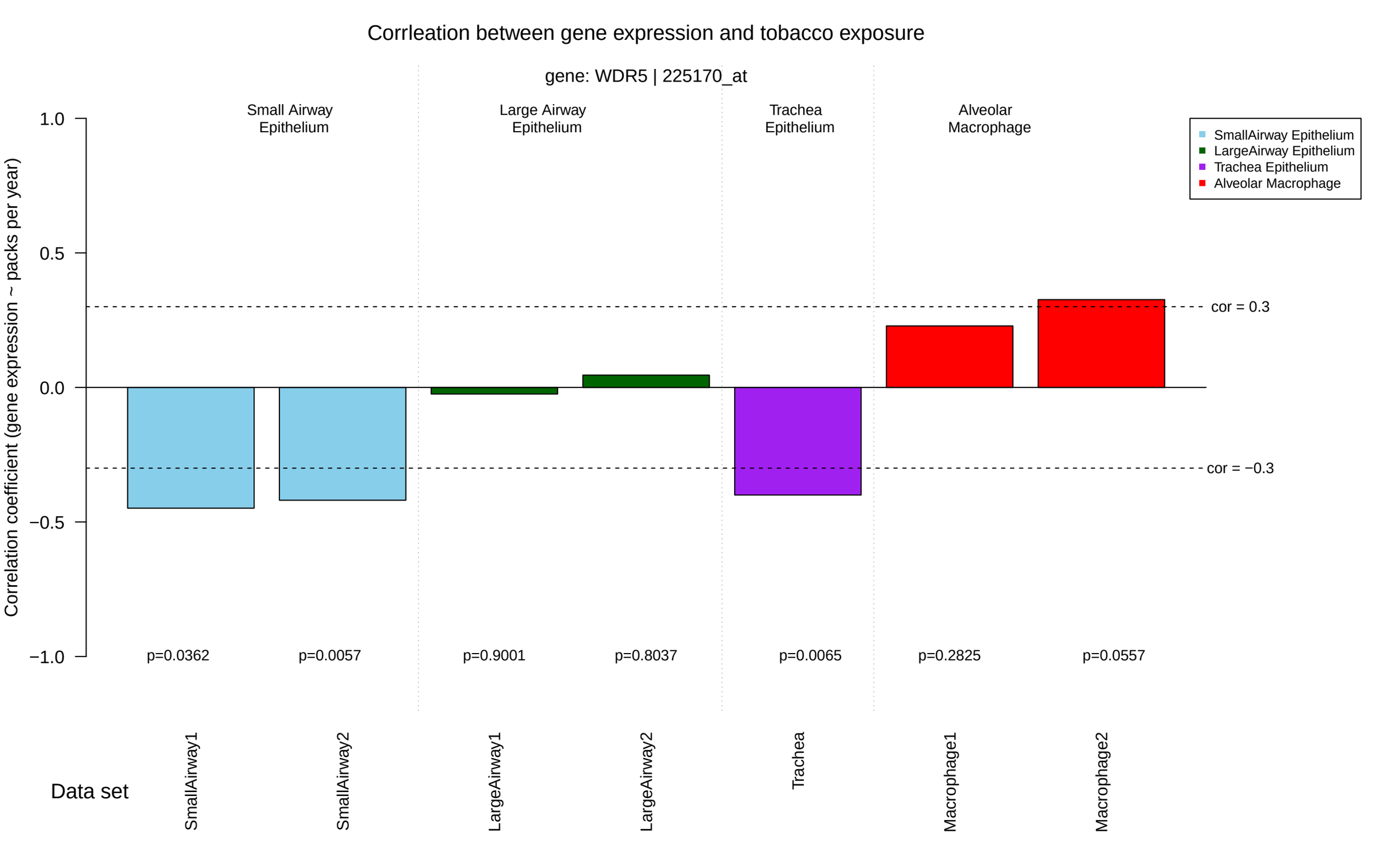

set: LargeAirway1 (p=0.0297)
WDR5 | 225170_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9618)
WDR5 | 225170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0164)
WDR5 | 225170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2872)
WDR5 | 225170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0238)
WDR5 | 225170_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WDR5 | 225170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0362
p=0.0057
p=0.9001
p=0.8037
p=0.0065
p=0.2825
p=0.0557
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2